

ANN MAHÉ

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EDUCATION

Columbia University, New York, NY

2027

B.S. Computer Science | GPA: 3.7 | Dean's List

Relevant Coursework: Advanced Programming, Data Structures & Algorithms, Computer Systems, Databases, Data Science, Quantum Computing, Artificial Intelligence, Databases, Human Centered Design

EXPERIENCE

Harmoniqs, Inc. – Software Engineering Intern

New York, NY / Jun – Aug 2026

- Owned a distributed trapped-ion simulation pipeline end to end, leading cross functional team of 3 to define and execute an 11 milestone roadmap and reproduce 4 published results on cloud infrastructure
- Cut cloud cost and runtime 5x by profiling memory bottlenecks and distributing 15 workloads across CPU and A100 GPU instances with fault-tolerant RAM-aware scheduling, raising usage 13% to 65%
- Shipped VS Code extension backed by AWS Bedrock that retrieves across internal docs and code to answer engineering questions in editor, adding chunking and citation logic to keep answers grounded
- Reverse engineered and implemented a parallel entangling gate algorithm verifying synthesized pulses at over 0.999 fidelity across two independent simulation engines, and scaling reliably to 7–45 ion chains

IonQ – Software Engineering Intern

New York, NY / Jun – Aug 2025

- Prototyped alerts and incident response automation for IonQ Aria's system control codebase, replacing fixed calibration schedules with telemetry thresholds and reducing overall system downtime by 50%
- Developed and optimized scalable ETL pipelines connecting InfluxDB NoSQL time series queries to Grafana dashboards, delivering real-time system monitoring and drift diagnostics across all 25 qubits
- Designed and developed modular object-oriented Python subsystems integrating time-series processing, statistical normalization, and correlation analyses of calibration outputs over large scale distributed data
- Worked cross-functionally with team of physicists to design, deploy, and improve reliability of software solutions, presented results from large-scale data to both technical and non-technical stakeholders

InfoPro Digital Limited – Machine Learning Intern

Paris, France / Jun – Aug 2024

- Collaborated directly with the Head of Product to determine user requirements, evaluate technical tradeoffs, and ship GenAI driven features into existing production ESG compliance software backend
- Built and deployed LLM/RAG microservices integrating Pinecone vector search REST APIs to summarize legal texts, improving retrieval and indexing accuracy and cutting retrieval times by 50%
- Designed and developed robust web scraper to automate content extraction, handling over 100 websites with Java-heavy frameworks and varying HTML layouts, and reducing current operational costs by 20%
- Delivered bi-weekly sprint updates to stakeholders, collaborating with five international content teams to incorporate feedback, account for tradeoffs, and ensure all project milestones were shipped on time

PROJECTS & RESEARCH

AptFinder | *Full Stack Project for my NYC Apartment Search* | Python, SQLite, Claude Code

2026

- Designed Python ingestion service and SQLite data model detecting and scoring listings in <4 minutes
- Automated listing extraction with GenAI APIs, ranked results by commute, walkability, and violations

Spatial and Probabilistic Influence in Cooperator-Defector Games | *Research Paper* | Python

2025

- Built algorithms combining game theory, computer science, & biophysics to simulate cell dynamics
- Published in the Biophysical Journal and presented findings at BPS Global Physics Summit in 2025

LEADERSHIP EXPERIENCE

CALIN Organization – Founder and Teacher

Shanghai, China / Jan 2018 - Present

W.M. Keck Physics Department – Physics Teaching Assistant

Claremont, CA / Jan 2024– May 2025

Claremont McKenna College – Resident Technology Assistant

Claremont, CA / Sep 2023 - May 2025

SKILLS, INTERESTS, AWARDS

- **Skills:** Python, C, SQL, R, PostgreSQL, Julia, Java, Unix/Linux, Matlab, CI/CD, REST APIs
- **Languages:** French (native), Mandarin (native), Spanish (C1 level)
- **Interests:** guzheng (Chinese harp), baking, snowboarding, innertube water polo, beach volleyball